
Cloudflare Radar

2024 Year in Review

Internet patterns and trends observed through Cloudflare's global network

Data-driven insights for technology executives and network infrastructure leaders

17%

Traffic Growth

3.3x

Starlink Surge

225

Disruptions

6.5%

Traffic Mitigated

17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet traffic grew 17% YoY; Starlink expanded satellite connectivity to underserved regions

+17%

YoY Global Traffic Growth

3.3x

Starlink Traffic Surge

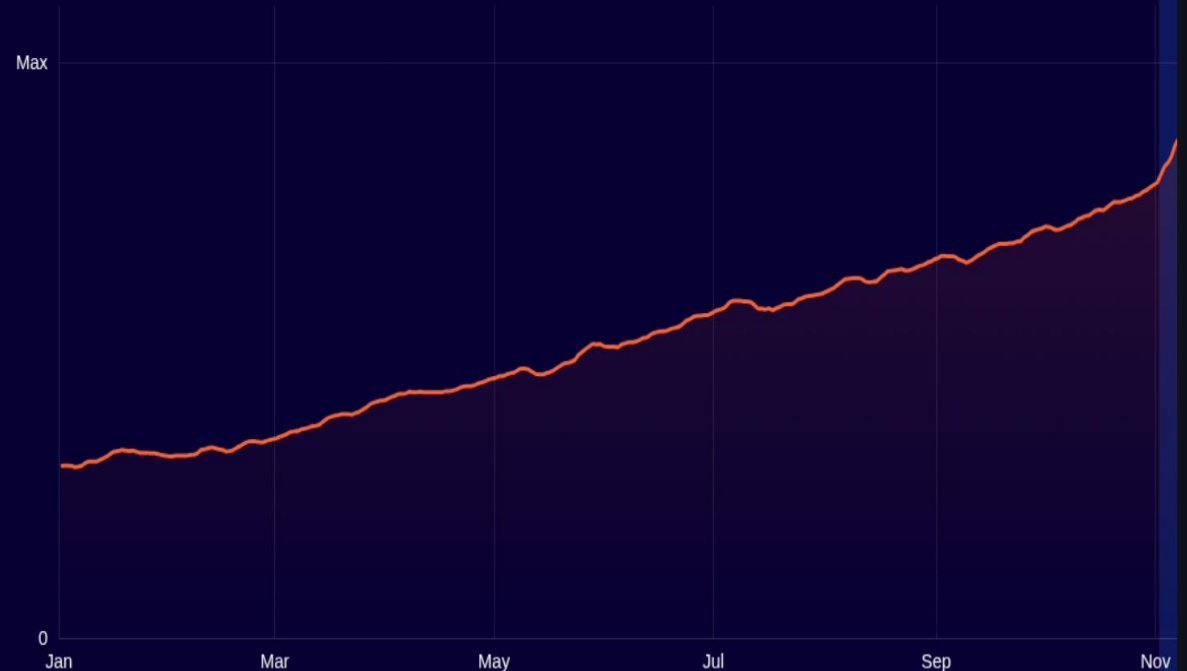
41%

Mobile Device Traffic

Global Traffic Growth Drivers

- Continuing upward trend driven by increasing reliance on Internet for communication, commerce, entertainment, and transportation
- Trend normalized from mid-January to account for post-holiday activity patterns
- SpaceX Starlink bringing satellite connectivity to previously underserved locations worldwide
- 41% of global traffic originated from mobile devices, reflecting critical mobile connectivity role

Normalized Starlink traffic worldwide



Google Leads Top 10 Services as Generative AI Rises

Ranking based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver

Top 10 Most Popular Internet Services

#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Key Observations

- Google maintained #1 for the second consecutive year
- Generative AI continued its rapid rise throughout 2024
- Metaverse receded into the background

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Technology adoption shifts across encryption, protocols, and developer ecosystem

2% → 13%

Post-Quantum TLS 1.3

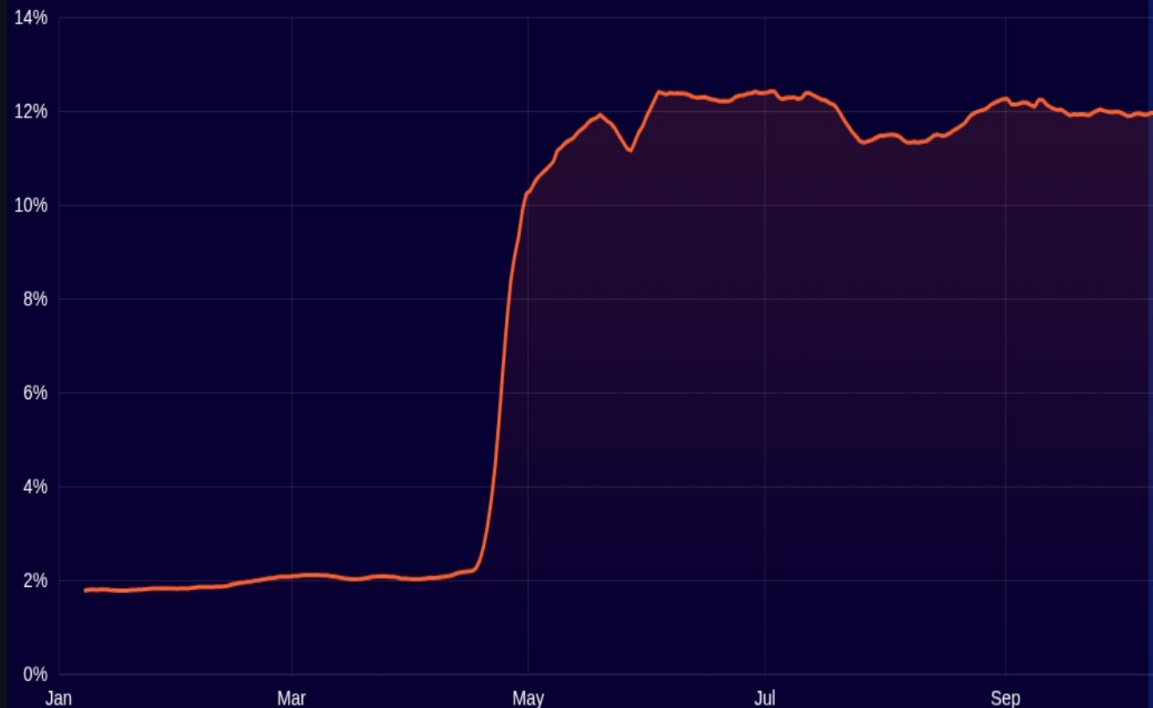
21%

HTTP/3 Adoption

~12%

Go — Top API Language

Post-quantum encrypted traffic worldwide



Post-Quantum Encryption

- Chrome 124 enabled PQ key exchange
- Adoption surged from ~2% to 13%
- Mozilla Firefox rolling out PQ I
- Apple Safari began initial post

Additional Metrics

- Search: Google 89% · Yandex 3.1% · Baidu 2.7% · Bing 2.6%
- Mobile OS: iOS 33% of traffic · Android peaks >95% in some regions
- Browsers: Chrome remains the most popular globally

API Client Language Share

Go	~12%
Node.js	—
Python	9.6%
Java	7.4%
.NET	3.6%

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Internet disruptions observed and tracked via Cloudflare Radar Outage Center

225

Total Disruptions

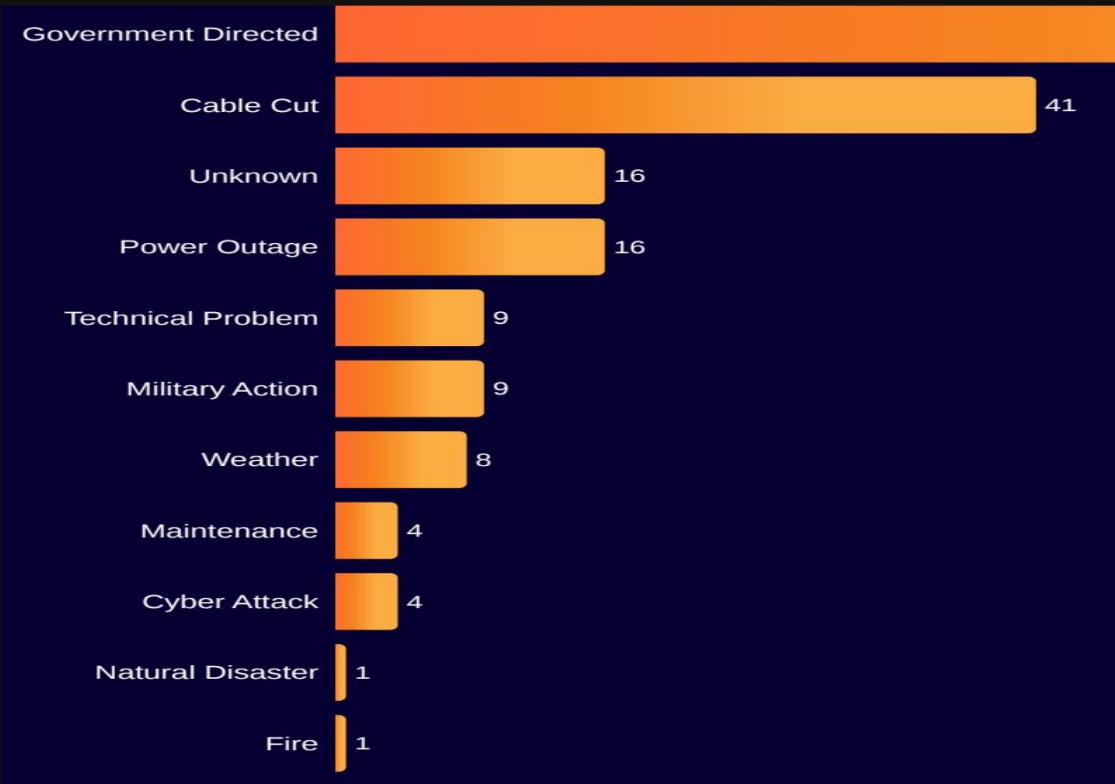
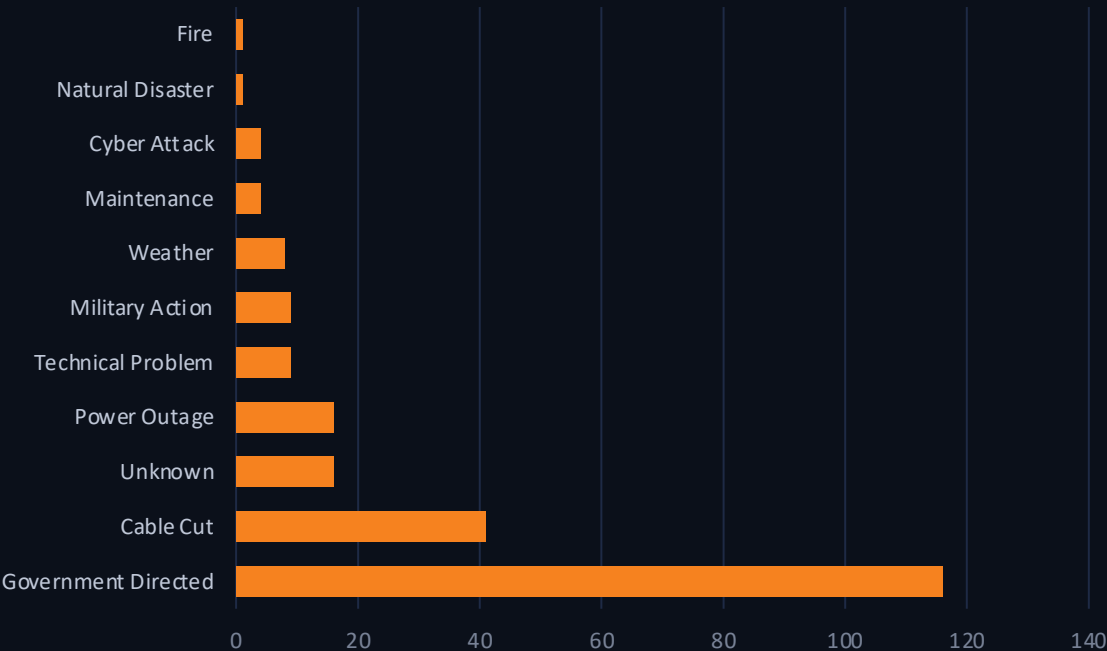
116+

Gov-Directed Shutdowns

41

Cable Cuts

Incidents



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Security findings from Cloudflare's global traffic analysis in 2024

6.5%

Traffic Mitigated

68%

Bot Traffic from Top
10 Countries

4.3%

Malicious Emails

+6.4%

RPKI Valid Routes

Threat Landscape

Most Attacked Sector: Gambling & Games

The Gambling/Games industry was the most targeted sector across Cloudflare's network in 2024.

Log4j (CVE-2021-44228) Still Exploited

First disclosed Dec 2021, Log4j continues to be actively targeted more than 2 years later, underscoring the persistent nature of widely known vulnerabilities.

Bot Traffic Concentration

68% of global bot traffic originated from just the top 10 countries, indicating geographic concentration of automated threat sources.

Email Security Threats

- 4.3% of all emails processed were identified as malicious
- 43% of malicious emails contained deceptive links
- .bar TLD originated the largest share of malicious & spam email
- Phishing via deceptive links remains the dominant email attack vector

Routing & Infrastructure Security

- RPKI valid routes increased by 6.4% — improving Internet routing security
- IPv6 adoption reached 28% of dual-stack traffic
- Spain recorded the highest download speed among all countries

Five Takeaways Shaping the 2025 Internet Outlook

Strategic insights from Cloudflare Radar's 2024 data for planning ahead

1

Traffic Growth Trajectory

Global Internet traffic grew 17% YoY with Starlink surging 3.3x — satellite connectivity is expanding rapidly to underserved regions, reshaping the global access landscape.

2

Post-Quantum Encryption Momentum

PQ encryption leapt from 2% to 13% of TLS 1.3 traffic after Chrome 124's default enablement. With Firefox and Safari following, enterprises must prepare cryptographic migration strategies.

3

Connectivity Fragility from Government Actions

Government-directed shutdowns were the leading cause of 225 major disruptions (116+ incidents), far exceeding infrastructure failures. Political risk is a primary connectivity risk.

4

Persistent and Evolving Security Threats

6.5% of traffic mitigated, Gambling/Games most targeted, Log4j still exploited 2+ years post-disclosure. Defense-in-depth and rapid patching remain essential.

5

Evolving Developer Ecosystem

Go became the top API client language (~12%). HTTP/3 reached 21% adoption. The developer toolchain is shifting — monitor ecosystem changes for infrastructure planning.